UNIT
8

BIOLOGICAL OXIDATION



Albert Lester Lehninger and Eugene Patrick Kennedy

Albert Lester Lehninger was an American Biochemist in the field of bioenergetics. He made fundamental contributions to the current understanding of metabolism at a molecular level. In 1948, he discovered with Eugene Patrick Kennedy (American Biochemist known for his work on lipid metabolism and membrane function) that mitochondria are the site of oxidative phosphorylation in eukaryotes, which introduced in the modern study of energy transduction.

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Learning Objectives

After studying this unit the students will be able to

- Explain the cellular respiration and how cells make ATP
- The basis of redox reaction and redox potential
- Structural insights of mitochondria
- Components of Electron transport chain
- Inhibitors of the electron transport chain
- The basis of oxidative phosphorylation
- ATP synthesis and mechanics of ATP synthase
- Uncouplers of oxidative phosphorylation

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